

Challenge 4: Solution

For Loop and While Loop

```
for (initialization; condition; update) {  
    // body  
}
```

```
while (condition) {  
    // body  
}
```

Sum of first N natural numbers using for loop

```
#include<stdio.h>  
void main()  
{  
    int n;  
    int i,sum=0;  
    printf("enter a number:");  
    scanf("%d",&n);  
    for(i=1;i<=n;i=i+1) //4<=3;i=4  
    {  
        sum=sum+i; //sum=3+3=6  
    }  
    printf("%d",sum);  
}
```

Sum of first N natural numbers using while loop

```
#include<stdio.h>  
void main()  
{  
    int n;  
    int i=1,sum=0;  
    printf("enter a number:");  
    scanf("%d",&n);  
    while(i<=n)  
    {  
        sum=sum+i;  
        i=i+1;  
    }  
    printf("%d",sum);  
}
```

Factorial of a number

```
#include<stdio.h>  
void main()  
{  
    int fact=1,n;  
    printf("enter a number:");  
    scanf("%d",&n);  
    if(n==0 || n==1)  
    {  
        printf("Factorial of %d is 1",n);  
    }
```

```

}
else
{
    for(int i=2;i<=n;i=i+1)
    {
        fact=fact*i;
    }
    printf("Factorial of %d is %d",n,fact);
}
}

```

Reverse a number

```

#include<stdio.h>
void main()
{
    int n=125,k;
    while(n!=0)
    {
        k=n%10;
        n=n/10;
        printf("%d",k);
    }
}

```

Reverse a number

```

#include<stdio.h>
void main()
{
    int n=125,k,sum=0;
    while(n!=0)
    {
        k=n%10;    //k=1
        sum=sum*10+k; //sum=521
        n=n/10;    //n=0
    }
    printf("%d",sum);
}

```

Palindrome or not

```

#include<stdio.h>
void main()
{
    int n=1221,sum=0;
    int c=n;
    while(n!=0)
    {
        sum=sum*10+n%10; //sum=121
        n=n/10;    //n=0
    }
    if(c==sum)
        printf("Palindrome");
}

```

```
else
printf("Not a palindrome");
}
```

Sum of digits

```
#include<stdio.h>
void main()
{
    int n=985,sum=0;
    int c=n;
    while(n!=0)
    {
        sum=sum+n%10; //sum=22
        n=n/10;      //n=0
    }
    printf("%d",sum);
}
```

Prime or not

```
#include <stdio.h>
void main()
{
    int num;
    int flag=0;
    printf("Enter a number:");
    scanf("%d",&num);
    for(int i=2;i<num;i=i+1)
    {
        if(num%i==0)
        {
            flag=1;
            break;
        }
    }
    if(flag==0)
    printf("\nIt is prime");
    else
    printf("It is not a prime number");
}
```

Prime or not using flag

```
#include <stdio.h>
void main()
{
    int num;
    int flag=0;
    printf("Enter a number:");
    scanf("%d",&num);
    for(int i=2;i<num;i=i+1)
    {
        if(num%i==0)
        {
            flag=1;

```

```

        break;
    }
}
if(flag==0)
printf("\nIt is prime");
else
printf("It is not a prime number");
}

```

Prime or not optimized code

```

#include <stdio.h>
void main()
{
    int num;
    int flag=0;
    printf("Enter a number:");
    scanf("%d",&num);
    for(int i=2;i<num/2;i=i+1)
    {
        if(num%i==0)
        {
            flag=1;
            break;
        }
    }
    if(flag==0)
    printf("\nIt is prime");
    else
    printf("It is not a prime number");
}

```

Challenge 4: Accepted



1. Even/Odd Numbers up to N

Write a C program to print all even and odd numbers from 1 up to a given number N.
Input: 10 → Output: Odd: 1 3 5 7 9 Even: 2 4 6 8 10

```

#include <stdio.h>
int main()
{

```

```

#include <stdio.h>
int main()
{

```

```

int n, i;
scanf("%d", &n);
printf("Odd: ");
for(i = 1; i <= n; i++)
{
    if(i % 2 != 0)
        printf("%d ", i);
}
printf("\nEven: ");
for(i = 1; i <= n; i++)
{
    if(i % 2 == 0)
        printf("%d ", i);
}
return 0;
}

```

```

int n, i;
scanf("%d", &n);
printf("Odd: ");
i = 1;
while(i <= n)
{
    if(i % 2 != 0)
        printf("%d ", i);
    i++;
}
printf("\nEven: ");
i = 1;
while(i <= n)
{
    if(i % 2 == 0)
        printf("%d ", i);
    i++;
}
return 0;
}

```

2. Count Digits in a Number

Write a C program to count the total number of digits in a given number.

Input: 45210 → Output: Number of digits = 5

```

#include <stdio.h>
int main()
{
    int n, count = 0, i;
    scanf("%d", &n);
    for(i = n; i > 0; i = i / 10)
        count++;
    printf("Number of digits = %d", count);
    return 0;
}

```

```

#include <stdio.h>
int main()
{
    int n, count = 0;
    scanf("%d", &n);
    while(n > 0)
    {
        count++;
        n = n / 10;
    }
    printf("Number of digits = %d", count);
    return 0;
}

```

3. Reverse a Number

Write a C program to reverse the digits of a given number.

Input: 1234 → Output: Reversed number = 4321

```

#include <stdio.h>
int main()
{
    int n, rev = 0, digit, i;
    scanf("%d", &n);
    for(i = n; i > 0; i = i / 10)
    {

```

```

#include <stdio.h>
int main()
{
    int n, rev = 0, digit;
    scanf("%d", &n);
    while(n > 0)
    {

```

```

    digit = i % 10;
    rev = rev * 10 + digit;
}
printf("Reversed number = %d", rev);
return 0;
}

```

```

    digit = n % 10;
    rev = rev * 10 + digit;
    n = n / 10;
}
printf("Reversed number = %d", rev);
return 0;
}

```

4. Sum of Digits

Write a C program to find the sum of all digits of a given number.

Input: 123 → Output: Sum of digits = 6

```

#include <stdio.h>
int main()
{
    int n, sum = 0, digit, i;
    scanf("%d", &n);
    for(i = n; i > 0; i = i / 10)
    {
        digit = i % 10;
        sum = sum + digit;
    }
    printf("Sum of digits = %d", sum);
    return 0;
}

```

```

#include <stdio.h>
int main()
{
    int n, sum = 0, digit;
    scanf("%d", &n);
    while(n > 0)
    {
        digit = n % 10;
        sum = sum + digit;
        n = n / 10;
    }
    printf("Sum of digits = %d", sum);
    return 0;
}

```

5. Armstrong Number Check

Write a C program to check whether a given 3-digit number is an Armstrong number.

Input: 153 → Output: 153 is an Armstrong number

```

#include <stdio.h>
int main()
{
    int n, i, digit, sum = 0;
    scanf("%d", &n);
    for(i = n; i > 0; i = i / 10)
    {
        digit = i % 10;
        sum = sum + digit * digit * digit;
    }
    if(sum == n)
        printf("%d is an Armstrong number", n);
    else
        printf("%d is not an Armstrong
number", n);
    return 0;
}

```

```

#include <stdio.h>
int main()
{
    int n, digit, sum = 0;
    scanf("%d", &n);
    while(n > 0)
    {
        digit = n % 10;
        sum = sum + digit * digit * digit;
        n = n / 10;
    }
    if(sum == 153)
        printf("153 is an Armstrong number");
    else
        printf("153 is not an Armstrong
number");
    return 0;
}

```

6. Power of a Number (a^b) using for/while loop

Write a C program to calculate the value of a number raised to a given power.

Input: Base = 2, Exponent = 5 → Output: 2⁵ = 32

```
#include <stdio.h>
int main()
{
    int base, exponent, i, result = 1;
    scanf("%d %d", &base, &exponent);
    for(i = 1; i <= exponent; i++)
    {
        result = result * base;
    }
    printf("%d^%d = %d", base, exponent,
result);
    return 0;
}
```

```
#include <stdio.h>
int main()
{
    int base, exponent, result = 1;
    scanf("%d %d", &base, &exponent);
    while(exponent > 0)
    {
        result = result * base;
        exponent--;
    }
    printf("%d^%d = %d", base, exponent,
result);
    return 0;
}
```

7. Palindrome Number Check

Write a C program to check whether a given number is a palindrome.

Input: 121 → Output: 121 is a palindrome

```
#include <stdio.h>
int main()
{
    int n, original, digit, rev = 0, i;
    scanf("%d", &n);
    original = n;
    for(i = n; i > 0; i = i / 10)
    {
        digit = i % 10;
        rev = rev * 10 + digit;
    }
    if(rev == original)
        printf("%d is a palindrome", original);
    else
        printf("%d is not a palindrome",
original);
    return 0;
}
```

```
#include <stdio.h>
int main()
{
    int n, original, digit, rev = 0;
    scanf("%d", &n);
    original = n;
    while(n > 0)
    {
        digit = n % 10;
        rev = rev * 10 + digit;
        n = n / 10;
    }
    if(rev == original)
        printf("%d is a palindrome", original);
    else
        printf("%d is not a palindrome",
original);
    return 0;
}
```

8. Display Cube of Numbers Up to an Integer

Write a program in C to display the cube of the number up to an integer.

Test Data :

Input number of terms : 5

Expected Output :

Number is : 1 and cube of the 1 is :1

Number is : 2 and cube of the 2 is :8

Number is : 3 and cube of the 3 is :27

Number is : 4 and cube of the 4 is :64
Number is : 5 and cube of the 5 is :125

```
#include <stdio.h>
int main()
{
    int n, i;
    printf("Input number of terms : ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        printf("Number is : %d and cube of
the %d is : %d\n",i, i, i * i * i);
    }
    return 0;
}
```

```
#include <stdio.h>
int main()
{
    int n, i = 1;
    printf("Input number of terms : ");
    scanf("%d", &n);
    while(i <= n)
    {
        printf("Number is : %d and cube of
the %d is : %d\n",i, i, i * i * i);
        i++;
    }
    return 0;
}
```